CASE STUDY

FlyRank: Content Refresh Opportunity Scoring

Lane 2 - Machine Learning Engineering Internship - Weeks 1-7

One-line outcome: A ranked queue that helps a content team with 20-50 pages per week of review capacity pick the right ones first - with a model that beats the transparent baseline by 3x on Precision@50.

The problem

A content team at a digital publisher has thousands of pages and maybe twenty hours a week to review them. Which pages deserve those twenty hours?

Pick wrong, and two things happen: you waste time rewriting a page that was already fine, or you miss a page that is quietly bleeding traffic. The second one hurts more - a high-impression page in decline has a shrinking window to recover.

I was two weeks into the FlyRank ML internship when I hit this. The assignment was not "build a model." It was: pick a lane, frame a question, and prove with real numbers that the lane is worth seven more weeks of your life.

What I did (and what I decided)

I read the full lane guide - sixteen sections, from data contracts to leakage audits - before I touched a line of code. Most people skip to the model. I did not. I needed to understand what "the right page to fix" actually means in this context: it is not a page guaranteed to recover. It is the page a reviewer should look at first, given limited capacity.

I chose Lane 2: Refresh / Content Opportunity Scoring. Not because it sounded impressive, but because it produces a ranked queue with reason codes - something a real person can act on Monday morning.

Quick look at the data

I pulled three real numbers from the starter dataset:

- 1,847 pages are both visible (500+ impressions in 90 days) and stale (6+ months old). That is 6.2% of the inventory sitting there with an audience but aging content.
- 2,103 pages are declining (trend = down) and have meaningful demand (100+ impressions). That is 7.0% losing visibility despite having search interest.
- 1,156 pages rank on page 1-2 (position 1-20) but have CTR under 0.5%. That is 3.9% under-capturing clicks they should be getting.

Model results

Method	ROC AUC	Avg Precision	Precision@50
Baseline rules	0.627	0.468	0.240
Logistic Regression	0.700	0.522	0.400

Decision Tree	0.742	0.575	0.540
Random Forest	0.750	0.618	0.740

Precision@50 asks: of the top 50 pages the system says to review first, how many actually turned out positive? The baseline got about 12 right. The random forest got about 37 right. That is strong evidence that a learned ranking can beat a fixed rule - on this starter slice.